# Statistics for International Commerce

## Week 15: Time Series and Forecasting

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# Agenda

- Part 1. Introduction to Time Series
- Part 2. Components of Time Series
- Part 3. Forecasting Methods
- Part 4. Forecast Accuracy
- Part 5. Python Example
- Part 6. Business Case

# Learning Objectives

After this lecture, you should be able to:

- explain what a time series is and distinguish it from cross-sectional data
- identify and interpret trend, seasonal, cyclical, and random components
- compute forecasts using naive, moving average, weighted moving average, and exponential smoothing methods
- evaluate and compare forecast performance using MAD and MSE
- implement and visualize a simple moving-average forecast in Python
- use forecasting results to support basic inventory planning decisions

# Part 1. Introduction to Time Series

# What Is a Time Series?

A time series is a sequence of observations collected over time.

Examples:

- monthly exports of Korea
- daily exchange rates
- quarterly GDP
- monthly company sales
- weekly website visitors

Question:

Why is time important here?

# Cross-Sectional vs Time Series Data

| Cross-sectional | Time series |
|---|---|
| Many units | One unit |
| One point in time | Multiple time periods |
| Countries in 2025 | Korea from 2010–2025 |

Examples:

Cross-sectional:

- GDP of 50 countries in 2025

Time series:

- Korea's GDP from 2000–2025

# Illustrative Example: Why Order Matters

Suppose weekly online sales are:

120, 90, 140, 100

If we ignore time order, we only see an average level.

If we keep time order, we can detect:

- ups and downs
- possible trend
- possible seasonal pattern

Key point: in time series, sequence carries information.

# Why Forecast?

Businesses use forecasts for:

- inventory planning
- staffing
- budgeting
- logistics
- production planning
- investment decisions

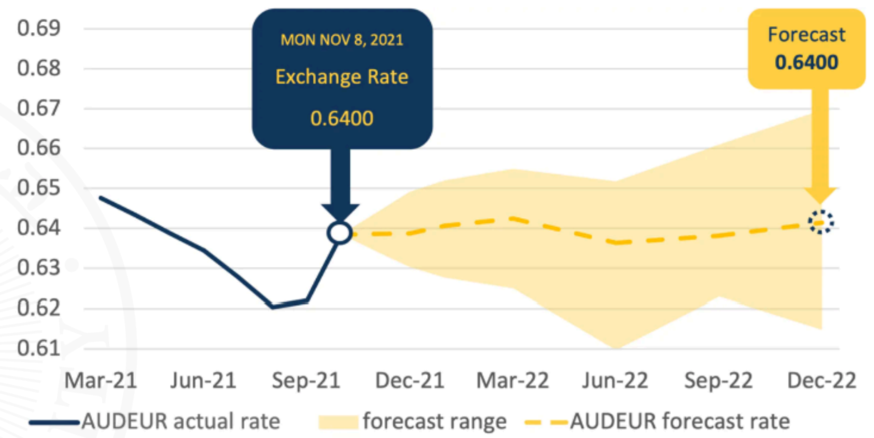
> To reduce uncertainty and improve decision-making.

# Forecast Examples

Forecast:

- exports
- imports
- shipping demand
- exchange rates
- tourism arrivals
- online sales

AUDEUR forecasts from bank majors 2023



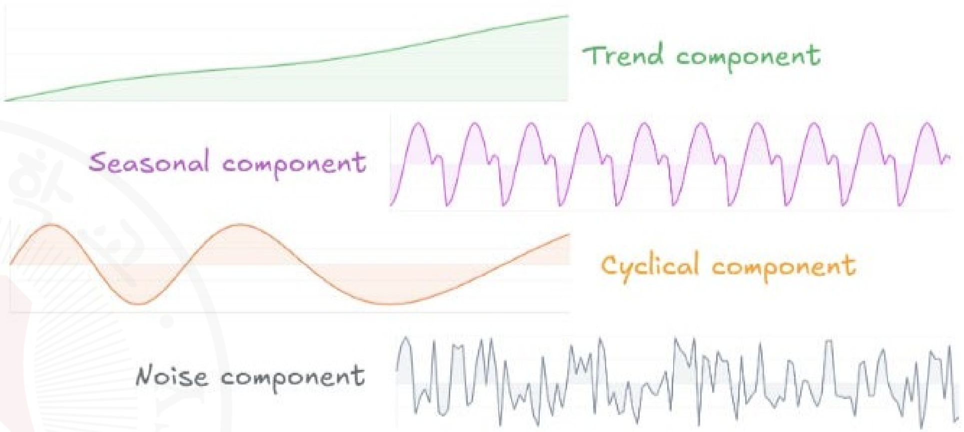

*Source: Multinational bank aggregate data*

# Part 2. Components of Time Series

# Four Components of Time Series

A time series may contain:

1. Trend
2. Seasonal variation
3. Cyclical variation
4. Random variation

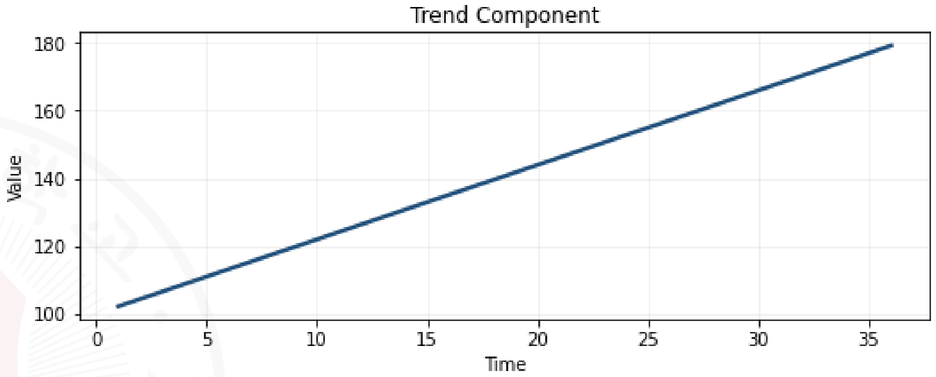


# Trend

Long-run movement.

Examples:

- growth in exports
- growth in e-commerce sales




> Key point: trend shows the overall direction of the series over time.

# Seasonality

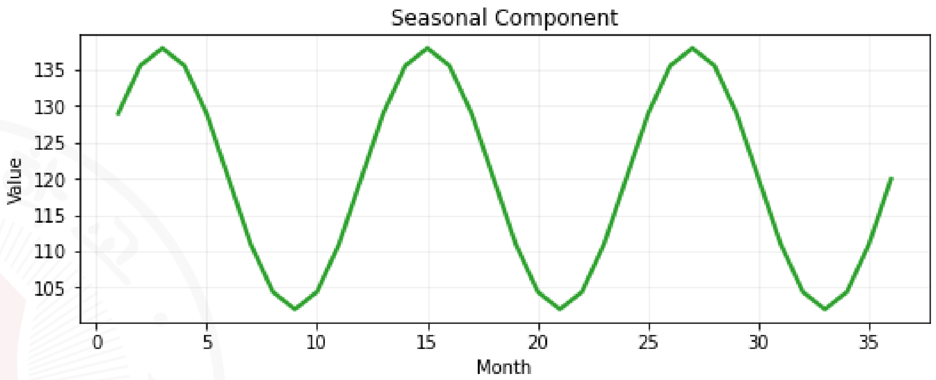
Pattern repeating regularly.

Examples:

- tourism demand
- holiday sales
- airline passengers

Example:

December sales often exceed February sales.



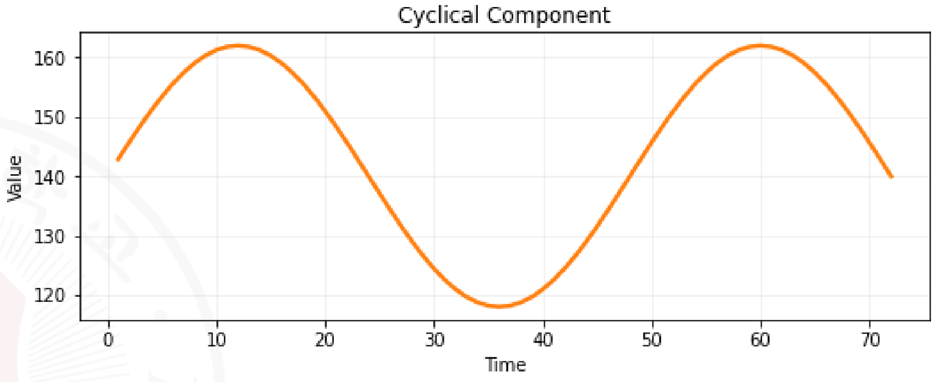

> Key point: seasonality reflects regular fluctuations tied to calendar periods.

# Cyclical Variation

Long-term business cycle effects.

Examples:

- recessions
- booms
- financial crises




> Key point: cyclical variation reflects economic expansions and contractions over several years.

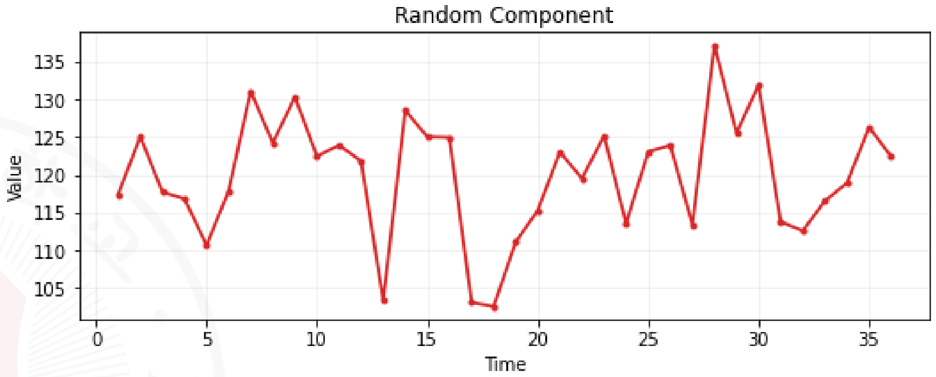
# Random Variation

Unexpected events.

Examples:

- natural disasters
- strikes
- pandemics

Cannot be predicted perfectly.




> Key point: random variation represents unpredictable fluctuations that cannot be systematically explained.

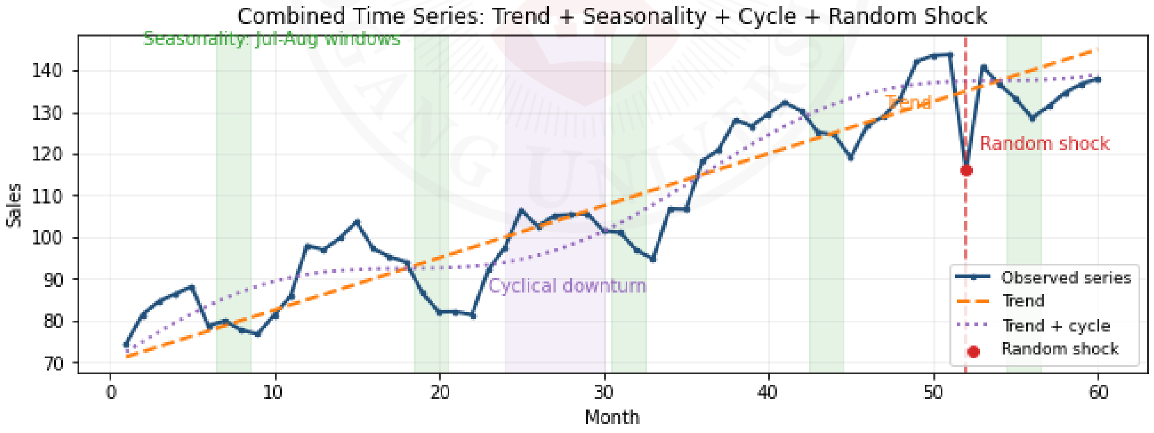
# Quick Component Identification Example

Monthly ice cream sales show:

- upward movement over 5 years
- peaks every July-August
- decline during recession years
- unusual drop during a pandemic month

Interpretation:

1. upward movement -> trend (because of growing demand)
2. July-August peaks -> seasonality (because of summer)
3. recession decline -> cyclical variation (because of economic cycle)
4. pandemic drop -> random variation (because of unexpected shock)



# Part 3. Forecasting Methods

# Naïve Forecast

Forecast next period using current period.

$$F_{t+1} = Y_t$$

Example:

Current month's sales:

[ 120 ]

Forecast next month:

[ 120 ]

> Key point: naïve forecast is simple but ignores trends and seasonality.

# Moving Average

Forecast uses several previous observations, where 'moving' refers to the window of past observations considered.

Example:

Sales:

110, 120, 130, 140, 170, ...

3-period moving average:

$$\frac{110+120+130}{3} = 120(\text{for month } 4); \frac{120+130+140}{3} = 130(\text{for month } 5)$$

> Key point: moving average smooths out short-term fluctuations but may lag behind trends.

# Practice

Monthly exports:

80, 90, 60, 110, 75, ...

Compute 3-month moving average forecasts for months 4, 5, and 6.

Answer:

$$(80 + 90 + 60)/3 = 76.67$$

$$(90 + 60 + 110)/3 = 86.67$$

$$(60 + 110 + 75)/3 = 81.67$$

# Why Moving Average?

Advantages:

- simple
- smooths fluctuations

Disadvantages:

- reacts slowly to sudden changes

# Weighted Moving Average

Recent observations receive higher weights.

Example:

Weights:

0.5, 0.3, 0.2

Observations:

130, 120, 110

Forecast:

$$0.5(130) + 0.3(120) + 0.2(110) = 123$$

> Key point: weighted moving average allows more flexibility by emphasizing recent data.

# Exponential Smoothing

Most popular business forecasting method.

Formula:

$$F_{t+1} = \alpha Y_t + (1 - \alpha) F_t$$

where:

$$0 < \alpha < 1 \text{ (smoothing constant, indicates weight on current observation)}$$

> Key point: exponential smoothing gives more weight to recent observations while still considering past forecasts.

## Interpretation of Alpha

Large α responds quickly; small α gives smoother forecasts.

# Example

$$\alpha = 0.30 \text{ (moderate weight on current observation)}$$

Current sales:

[ 100 ]

Previous forecast:

[ 90 ]

Forecast:

$$0.30(100) + 0.70(90) = 93$$

# Illustrative Example: Choosing Alpha

Assume:

- current actual demand $Y_t = 120$
- previous forecast $F_t = 100$

If $\alpha = 0.8$:

$$F_{t+1} = 0.8(120) + 0.2(100) = 116$$

If $\alpha = 0.2$:

$$F_{t+1} = 0.2(120) + 0.8(100) = 104$$

Higher $\alpha$ reacts faster to new information.

# Some other forecasting methods:

- Holt's linear trend method (for series with trend)
- Holt-Winters method (for series with trend and seasonality)
- ARIMA/SARIMA models (for complex patterns and autocorrelation)
- Machine learning methods (for large datasets and nonlinear relationships)

# Part 4. Forecast Accuracy

# Forecast Errors

Forecast error:

$$e_t = Y_t - F_t$$

Example:

- Actual: 100
- Forecast: 90
- Error: 10

```
## (80.0, 110.0)

## (-20.0, 20.0)
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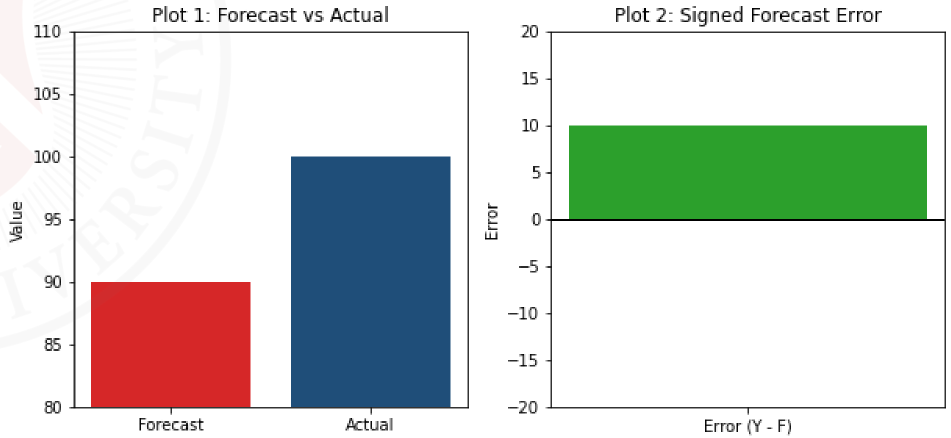


> Positive error means under-forecasting; negative error means over-forecasting.

# Mean Absolute Deviation (MAD)

$$\text{MAD} = \frac{\sum |e_t|}{n}$$

Measures average forecasting error.

Smaller is better.

> Key point: MAD treats all errors equally regardless of direction.

# Mean Squared Error (MSE)

$$\text{MSE} = \frac{\sum e_t^2}{n}$$

Large errors receive heavier penalties.

> Key point: MSE emphasizes larger errors more than MAD.

**Choose the method with smaller MAD and smaller MSE.**

# Worked Accuracy Example

Actual demand for 4 months: 100, 110, 120, 130

Method A

- Forecast: 98, 112, 118, 128
- Errors: 2, 2, 2, 2
- MAD = 2.0
- MSE = 4.0

Method B

- Forecast: 90, 105, 130, 125
- Errors: 10, 5, 10, 5
- MAD = 7.5
- MSE = 62.5

Conclusion: Method A is more accurate because it has the smaller MAD and MSE.

# Some other forecast accuracy measures:

- Mean Absolute Percentage Error (MAPE)

$$\text{MAPE} = \frac{100\%}{n} \sum \left| \frac{e_t}{Y_t} \right|$$

- Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{frac \sum e_t^2 n}$$

# Part 5. Python Example

# Why Python Here?

Python helps us turn the lecture ideas into a small forecasting workflow:

- build a dataset
- visualize the series
- create forecasts
- compare forecast accuracy

# Example Dataset and Trend

Monthly export demand shows a steady upward pattern.

| Month | Exports |
|---|---|
| 1 | 120 |
| 2 | 124 |
| 3 | 128 |
| 4 | 131 |
| 5 | 135 |
| 6 | 140 |
| 7 | 144 |
| 8 | 149 |
| 9 | 153 |
| 10 | 158 |
| 11 | 162 |
| 12 | 167 |

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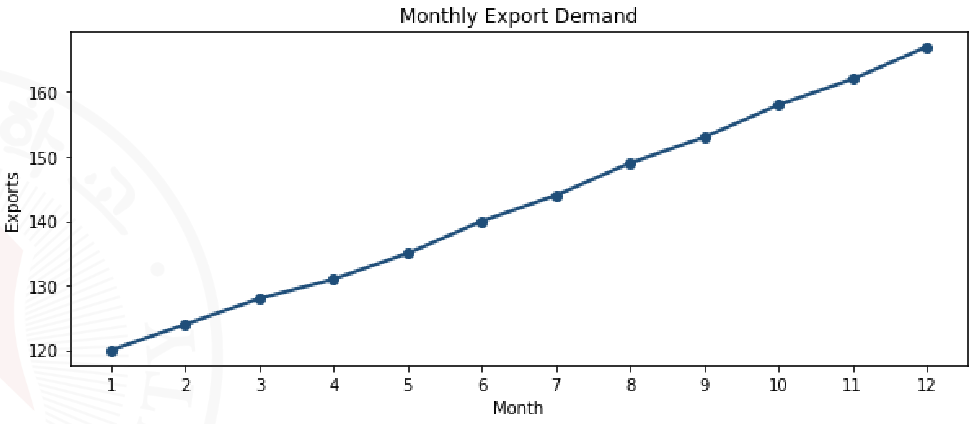


Interpretation:

- the level is rising over time
- month-to-month changes are small
- this is a good setting for comparing simple forecasting methods

> The line slopes upward, which is why simple forecasting methods are useful here.

# Forecast Output

The Python workflow produces two forecasts:

| Month | Actual | Naive | MA(3) |
|---|---|---|---|
| 4 | 131 | 128.0 | 124.0 |
| 5 | 135 | 131.0 | 127.7 |
| 6 | 140 | 135.0 | 131.3 |
| 7 | 144 | 140.0 | 135.3 |
| 8 | 149 | 144.0 | 139.7 |
| 9 | 153 | 149.0 | 144.3 |
| 10 | 158 | 153.0 | 148.7 |
| 11 | 162 | 158.0 | 153.3 |
| 12 | 167 | 162.0 | 157.7 |

> The chart makes the lag visible at a glance.

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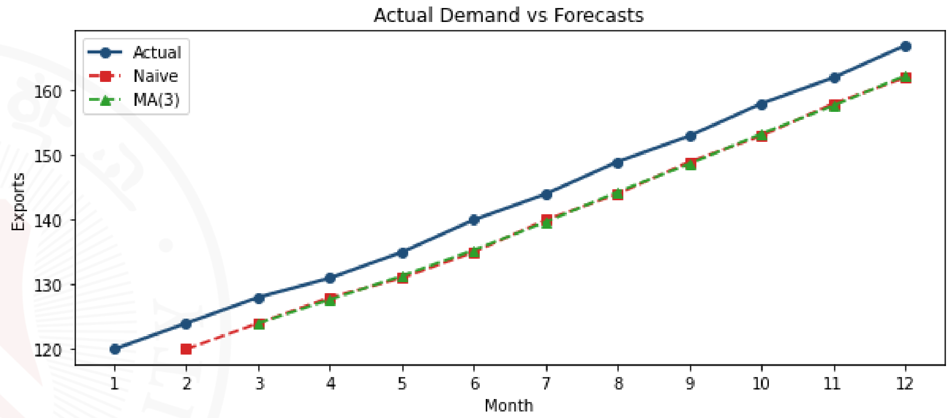


Interpretation:

- the naive forecast follows the most recent value
- MA(3) is smoother, but it lags behind the rising trend

# Accuracy Summary

| Method | MAD | MSE |
|---|---|---|
| Naive | 4.27 | 18.64 |
| MA(3) | 8.56 | 73.82 |

```
## (array([0, 1]), [Text(0, 0, 'Naive'), Text(1, 0, 'MA3')])
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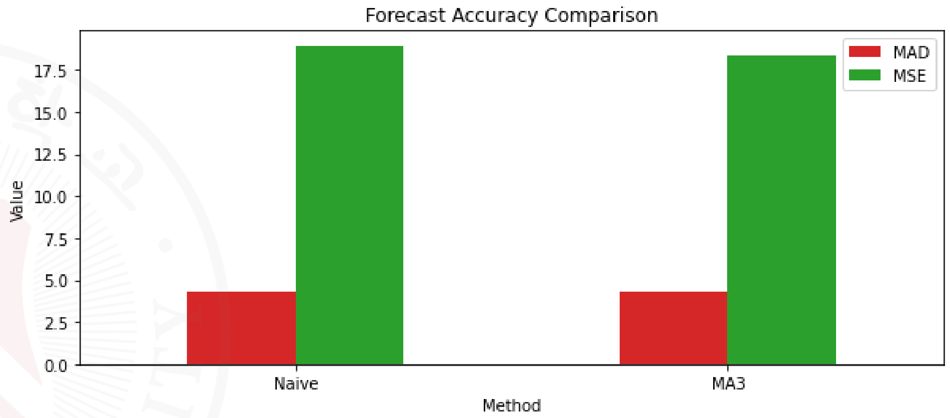


Conclusion:

- for this steadily rising series, the naive forecast is more accurate numerically
- MA(3) is still useful because it smooths short-term fluctuations
- MAD and MSE make the comparison objective and easy to explain

> Lower bars are better, and both MAD and MSE favor the naive forecast here.

# What to Remember

The key lesson is not only how to code the forecast, but how to read the output:

- tables show the forecast values directly
- figures would show the trend and the lag visually
- accuracy measures tell us which method fits the data better

# Part 6. Business Case

# Case Study

A Korean exporter observes monthly export demand:

| Month | Orders |
|---|---|
| Jan | 120 |
| Feb | 130 |
| Mar | 140 |
| Apr | 145 |
| May | 160 |

Tasks:

1. Calculate 3-month moving average.
2. Forecast June demand.
3. Recommend inventory planning.

# What We Learned

Time series components:

- trend
- seasonality
- cycles
- randomness

Forecasting methods:

- naïve
- moving average
- weighted moving average
- exponential smoothing

Forecast evaluation:

- MAD
- MSE

# Final Message

Forecasts are never perfect.

Good forecasting reduces uncertainty and improves business decision-making.

# Remaining Schedule

*(June 9)* Multiple Time Series and Forecasting (LMW Chapter 16)

- Final review and Q&A session

*(June 11)* Project presentations

*(June 16)* No class (final exam week)

*(June 18)* Final Exam

Any questions?

Thank you for your attention and active participation!